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Natural Recovery of Rough Fescue (*Festuca hallii* (Vasey) Piper) Grassland after Disturbance by Pipeline Construction in Central Alberta, Canada

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ABSTRACT: Many attempts to restore grasslands after pipeline construction, including soil manipulation such as topsoil stripping and replacement, and revegetation methods such as seeding native species or non-native cover species, have been unsuccessful. Recent pipeline techniques have used minimum disturbance and revegetation via natural recovery. The goal of this research was to evaluate natural recovery in rough fescue (*Festuca hallii* (Vasey) Piper) grassland in central Alberta, Canada. Plant species cover was evaluated, comparing pipeline right-of-way disturbances to undisturbed controls. Pipeline construction methods were assessed to determine which would be most successful in returning native grassland species. Pipelines with the most intense disturbances were dominated by wheatgrasses, with abundant bare ground and sparse moss and lichen cover; whereas those with the least intense disturbances had cover similar to native grassland. Results confirmed the importance of minimum disturbance in grasslands. Retaining grassland sod through plow-in pipeline construction and keeping disturbance as narrow as possible is critical to successful recovery. Rough fescue appears to recover better with plow-in pipeline construction than with seeding, most likely from intact sod. Therefore, narrow trenching with plow-in techniques is recommended for grasslands.

Index terms: *Festuca hallii*, grasslands, minimum disturbance, natural recovery, pipeline construction, rough fescue

INTRODUCTION

Grassland recovery after industrial disturbance that removes all vegetation and soil is highly complex. Pipeline construction strips vegetation, leaving subsoil on the surface, admixing soil horizons, compacting soils, and destroying biological crusts including moss and lichen (Naeth et al. 1987; Lovich and Bainbridge 1999). Resulting disturbed land is, in turn, subjected to weather, increasing susceptibility to runoff and erosion, faunal disturbance, and invasion by non-native plant species.

Rough fescue grassland is an important forage source in western Canada and a rapidly diminishing natural resource (Romo et al. 1990; Clark 1998). Although efforts have been directed to its restoration after disturbance, they have been mostly unsuccessful. Attempts to assist pipeline restoration in fescue grasslands included soil manipulation during construction, such as topsoil stripping and replacement, and revegetation methods, such as seeding native species or non-native cover crops. Seeding native rough fescue (*Festuca hallii* (Vasey) Piper or *Festuca campestris* Rydb.) has had no documented success. Desserud et al. (2010) and Elsinger (2009) reported numerous examples of unsuccessful revegetation of rough fescue following seeding on pipelines and well sites in central and southern Alberta.

Recent pipeline construction has used

minimum disturbance techniques with plow-in and narrow ditch pipelines, 60- to 80-cm wide disturbances (excavation bucket width), and sites left to revegetate via natural recovery. In central Alberta, Petherbridge (2000) found pipeline construction with no topsoil stripping resulted in a smaller initial disturbance and conserved more of the initial plant community than conventional right-of-way (ROW) stripping, although admixing reduced emergence of some native species. Time required for perennial establishment may expose the site to erosion and non-native species (Hammermeister and Naeth 1996). Some researchers commented that while initial species to emerge during natural recovery are undesirable forbs, slower growing perennials, such as rough fescue, will eventually dominate (Naeth et al. 1997; Arychuk 2001; Hammermeister 2001). Lathrop and Archbold (1980) concluded time since disturbance was a positive determinant for linear disturbance recovery, but extent of disturbance had a greater negative impact.

In this study, pipeline ROWs constructed in undisturbed grasslands and left to natural recovery were evaluated. Undisturbed adjacent grassland controls were compared to ROW species to assess effects of pipeline construction methods by examining vegetation cover and biological crust conditions. The objective was to assess which construction methods would result in returning native grassland species.

Methods

We conducted our study in the Rumsey Natural Area, a provincially protected area (182 km²) of uncultivated natural grasslands in the Central Parkland natural region of Alberta, Canada (between latitudes 51.796° and 51.883° and longitudes 112.417° and 112.701°; elevation: 900 m). The natural area is protected from residential and rural development; however, cattle grazing and oil and gas construction is allowed. Over 200 natural gas wells and pipelines have been constructed in the Rumsey Natural Area since the 1950s. Temperatures range from −40 to +35 °C with growing season (May to October) temperatures averaging 13 °C. Average annual rainfall is approximately 350 mm and snowfall 100 cm. Topography is undulating with a complex of small depressions and hills with loamy Dark Brown Chernozem soils on medium textured glacial till. Native vegetation on uplands and upper slopes is rough fescue grassland dominated by *F. hallii*, *Hesperostipa curtiseta* (A.S. Hitchc.) Barkworth (western porcupine grass), *Koeleria macrantha* (Ledeb.) J.A. Schultes (June grass), *Poa* spp. (native bluegrasses), and wheatgrasses: *Elymus lanceolatus* (Scribn. & J.G. Sm.) Gould ssp. *lanceolatus* (northern wheatgrass), *Pascopyron smithii* (Rydb.) A. Löve (western wheatgrass), and *Elymus trachycaulus* (Link) Gould ex Shinnars ssp. *subsecundus* (Link) A. & D. Löve (slender wheatgrass). Non-native species occurring on some disturbed sites include *Poa pratensis* L. (Kentucky bluegrass), *Cirsium arvense* (L.) Scop. (Canada thistle), and *Taraxacum officinale* G.H. Weber ex Wiggers (dandelion).

We studied 13 natural gas pipeline ROW segments between 2006 and 2010. Pipelines were selected with a stratified random process. We selected only pipelines left to natural recovery situated in undisturbed rough fescue grassland. Within these criteria, a ROW segment was randomly chosen on each pipeline for vegetation sampling.

Five pipelines were installed using a plow-in technique (Table 1). A plow creates a narrow trench the width of the bucket (80 cm), pipe (15 cm diameter) is fed into

Table 1. ROW sites showing year built, plant community classification, and construction method. Construction method is explained in the text.

Pipeline	Year Built	Plant Community	Construction Method
PA330	2009-2	Bluegrass-Fescue	Plow-in
PA1024	2009-1	Bluegrass-Fescue	Plow-in
PA65	2008	Bluegrass-Fescue	Plow-in
P991	1999-1	Bluegrass-Fescue	Plow-in
P992	1999-2	Bluegrass-Fescue	Plow-in
P901	1990-1	Bluegrass-Fescue	Topsoil-strip
P85	1985	Bluegrass-Fescue	Topsoil-strip
P00	2000	Wheatgrass	Topsoil-strip
P951	1995-1	Wheatgrass	Topsoil-strip
P952	1995-2	Wheatgrass	Topsoil-strip
P87	1990-2	Wheatgrass	Topsoil-strip
P902	1987	Wheatgrass	Ditch-witch
P83	1983	Wheatgrass	Ditch-witch

the trench the same day, and soil and sod are allowed to fall back into place. Six pipelines were topsoil-strip where topsoil was stripped the width of the trench (80 cm), stored, and replaced following pipe installation, usually within one week (Table 1). Two pipelines employed ditch-witch construction, whereby a trencher chops sod, mixing it with trench soil and causing a narrow, major disturbance, approximately 80 cm wide. Pipe is installed in the trench the same day, the broken sod/soil mix covering the pipe (Table 1). All ROWs were left to natural recovery (not seeded).

On each pipeline ROW, a 30-m transect was positioned along the trench, the area of highest disturbance, and another 15 m from the ROW in undisturbed grassland for a control. On each transect, ten 20-cm by 50-cm rectangular quadrats (Daubenmire

1959), 3 m apart, were used to visually assess foliar cover of all species, litter, and bare ground (Figure 1).

Student t-tests were used for comparisons between ROW and control plots. Pearson product correlation and linear regression were used to display relationships between species and ROW characteristics. Two-way cluster analysis with Ward’s method was used to classify ROW and control plots with a cut-off at 50% of information remaining. Cluster analysis was based on grouping points representing individuals with similar characteristics in mathematical space (Kent and Coker 1992). Non-metric multidimensional scaling (NMS) was used to ordinate and display the separation of native grassland controls to ROW species and to confirm cluster analysis groupings (McCune and Grace 2002). Key species

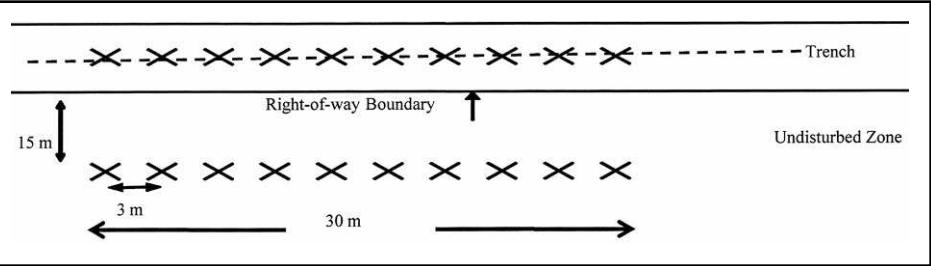


Figure 1. Sampling design (not to scale) showing pipeline right-of-way (ROW) boundary, 10 sampling points along the trench and 10 sampling points 15 m away from the ROW in undisturbed grassland.

and significant factors, such as bare ground, were overlain on NMS diagrams to display trends. Sorenson (Bray-Curtis) distance measure was used to calculate statistical significance of axes with a final run of 500 iterations using three axes and a randomly selected starting configuration. Indicator species analysis (ISA) validated dominant species of plant communities derived from cluster analysis groupings. Indicator values (IV) ranged from 0 to 100, where 100 indicated a species was exclusively found in a particular group (Dufrene and Legendre 1997). Nonparametric multiple response permutation procedure (MRPP), operating on Sorenson (Bray-Curtis) distance measures, was used to evaluate significant differences between ROW groups and controls (Zimmerman et al. 1985). MRPP generates a chance corrected within-group agreement value (A), which evaluates the difference between species composition of grouped plots. The lower the A value, the more similar are the groups (McCune and Grace 2002). PC-ORD (ver. 5.31, MjM Software, Gleneden Beach, OR) was used to classify, ordinate, and perform ISA and MRPP calculations, and PASW (ver. 18.0, SPSS, Chicago IL) and Excel (ver. 2007, Microsoft, Redmond WA) for statistical analyses.

RESULTS

Two plant communities were isolated on the ROWs through cluster analysis (Figure 2). One was dominated by *E. lanceolatus* and *P. smithii* (Wheatgrass community) and the other by *Poa* spp. and *F. hallii* (Fescue-Bluegrass community) (Table 2). Pipeline construction techniques affected *F. hallii* cover, which was significantly greater on plow-in pipelines, and wheatgrass cover, which was significantly greater on ditch-witch and topsoil-strip pipelines (Figure 3). Seventy percent of Fescue-Bluegrass ROWs were plow-in pipelines. All Wheatgrass community ROWs were topsoil-strip or ditch-witch pipelines (Table 1). *Festuca hallii*, moss and lichen, and litter cover were significantly greater on controls than ROWs (Figure 4). Wheatgrasses (*Elymus lanceolatus*, *E. trachycaulus*, and *P. smithii*) and bare ground cover were significantly lower on controls than ROWs (Figure 4).

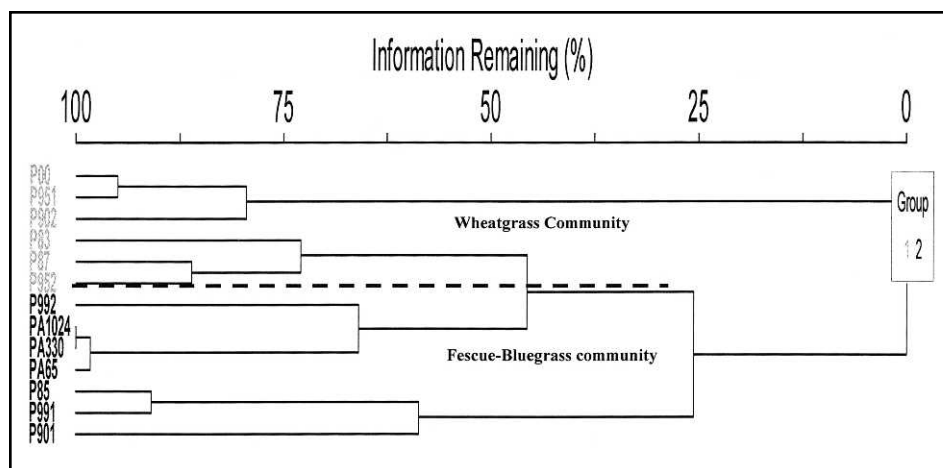


Table 2. ROW's grouped by dominant species and corresponding controls, mean percent cover (standard deviation), and differences between ROW and control groups tested by Indicator Species Analyses (IV) and t-tests of significance. Only species with greater than 1% cover are shown.

	Wheatgrass Community (n=6)					Fescue-Bluegrass Community (n=7)				
	ROW		Control			ROW		Control		
	Mean	IV	Mean	IV	P	Mean	IV	Mean	IV	P
<i>Carex</i> spp.	7.4(4)	49	7.8(2)	51	0.870	2.4(3)	26	2.9(4)	23	0.784
<i>Elymus lanceolatus</i>	15(10)	78	4.4(4)	22	0.035	3.1(2)	67	0.9(1)	19	0.016
<i>Elymus trachycaulus</i>	3.3(4)	69	0.7(1)	11	0.147	3.0(4)	70	0.7(1)	11	0.174
<i>Festuca hallii</i>	2.6(2)	12	16(7)	86	0.002	8.9(6)	78	27(20)	75	0.044
<i>Hesperostipa curtiseta</i>	7.6(7)	26	17(7)	69	0.037	8.9(7)	46	7.8(6)	47	0.769
<i>Koeleria macrantha</i>	6.9(4)	35	9.5(4)	58	0.327	2.0(2)	36	2.0(2)	43	0.992
<i>Pascopyrum smithii</i>	13(12)	59	1.7(3)	4	0.050	5.1(5)	73	0.9(2)	11	0.077
<i>Poa pratensis</i>	1.8(3)	32	1.0(2)	12	0.568	5.6(6)	14	1.6(3)	1	0.134
<i>Poa</i> spp.	3.4(4)	30	2.2(2)	33	0.511	7.8(6)	79	2.1(3)	15	0.052
Total grasses	67(7)	n/a	68(9)	n/a	0.913	47(22)	n/a	52(29)	n/a	0.799
<i>Achillea millefolium</i>	1.0(1)	43	0.6(1)	24	0.450	1.7(2)	52	1.1(1)	34	0.424
<i>Artemisia frigida</i>	16(8)	59	11(5)	41	0.232	4.5(3)	60	3.0(4)	29	0.463
Total forbs	32(7)	n/a	32(8)	n/a	0.961	23(14)	n/a	21(13)	n/a	0.695
<i>Rosa arkansana</i>	1.0(1)	14	2.5(3)	48	0.280	2.3(2)	63	0.8(1)	19	0.077
<i>Symphoricarpos occidentalis</i>	1.0(2)	10	0.6(1)	13	0.746	7.9(8)	49	3.6(5)	18	0.250
Total shrubs	2.0(3)	n/a	3.4(3)	n/a	0.397	10(8)	n/a	4.4(5)	n/a	0.131
Moss/lichen	3.3(3)	15	15(5)	82	0.020	1.9(3)	21	3.3(3)	46	0.438
Native species	107.0(8)	n/a	112.8(4)	n/a	0.807	86.1(40)	n/a	80.7(40)	n/a	0.149
Non-native species	5.6(5)	n/a	1.7(2)	n/a	0.055	9.6(8)	n/a	2.7(3)	n/a	0.110
Bare ground	14(15)	n/a	1.5(2)	n/a	0.064	26(24)	n/a	2.3(3)	n/a	0.025
Litter	14(3)	n/a	16(4)	n/a	0.370	19(11)	n/a	43(29)	n/a	0.068
Diversity	2.4(0)	n/a	2.6(0)	n/a	0.930	2.4(0)	n/a	2.3(0)	n/a	0.699
Richness	22(6)	n/a	27(4)	n/a	0.167	25(8)	n/a	25(5)	n/a	0.951

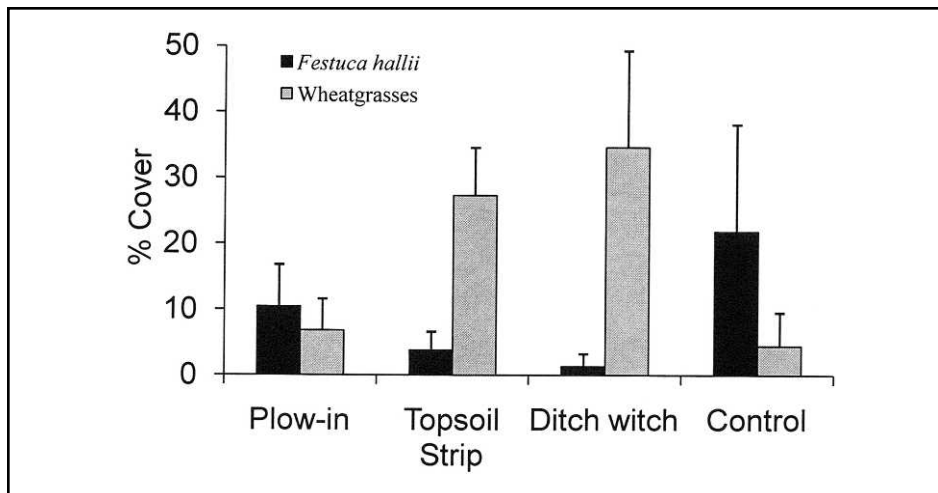


Figure 3. *Festuca hallii* and wheatgrass (*Elymus lanceolatus*, *Elymus trachycaulus*, and *Pascopyron smithii*) cover showing pipeline construction techniques and undisturbed controls. *Festuca hallii* ($p < 0.001$) decreased with degree of disturbance while wheatgrasses ($p = 0.001$) increased. Controls had greater *F. hallii* cover and less wheatgrass.

natural recovery resulted in moderate erosion and high diversity with mid-seral species, despite a slow progression towards pre-disturbance status. Soil exposure following disturbance may allow seed bank propagules to become established.

Rough fescue grassland species occurred on over half of our sites and similarity between fescue-bluegrass-dominated communities and controls indicated success of natural recovery. Pipeline construction techniques and degree of disturbance

also affected natural recovery success. In our study, plow-in pipeline construction reduced disturbance intensity with retention of much of the original sod, including viable perennial grass roots. Plow-in pipelines had the fewest non-native species, greatest rough fescue recovery, and were similar to native grassland. Maki (1989) also found plow-in construction avoided induced germination of non-native species, and other researchers noted plow-in pipelines had comparable species richness to undisturbed prairie when revegetated

by natural recovery (Petherbridge 2000; Elsinger 2009; Desserud et al. 2010).

Wheatgrass-dominated pipelines were associated with the greatest disturbance in our study – top-soil-strip or ditch-witch construction techniques. The disturbance extent and ensuing bare ground would provide suitable habitat for wheatgrass establishment, having high germination rates, rapid establishment, and rapid ground cover. Our findings are similar to Hammermeister (2001), who suggested aggressive wheatgrasses may suppress establishment of other species, and to Desserud et al. (2010), who observed wheatgrasses dominated top-soil-stripped pipelines. Ostermann (2001), who found rhizomatous non-native and native grasses dominated trenches of 12-year old pipelines in mixed grass prairie and foothills grasslands, suggested wheatgrasses got an early start and remained dominant. Elsinger (2009) observed ditch-witch construction left a poor growth medium for plants to establish through natural recovery.

Litter, a critical grassland component, was reduced on all ROWs compared to undisturbed grassland. Plant litter helps conserve soil water by reducing soil temperature and evaporation, and insulating soil against incident radiation by reducing light and temperature at the soil surface (Hopkins 1954; Johnston et al. 1971; Willms et al. 1985; Willms et al. 1986). Litter may have assisted plant community development in Fescue-Bluegrass ROWs in our study by regulating soil water, as Willms et al. (1986) and Foster and Gross (1998) suggest, and by providing micro-climates for various seeds as found by Molofsky and Augspyrger (1992). Conversely, lack of litter on Wheatgrass ROWs may have impeded plant community development. *Festuca hallii* recovery on ROWs with greater litter in our study is consistent with findings by Willms (1988), who observed reduced rough fescue growth when litter was removed.

Despite evidence of grassland recovery on some ROWs, bare ground was apparent on all ROWs. This is consistent with findings by Hammermeister (2001), who found bare ground was significantly greater on natural

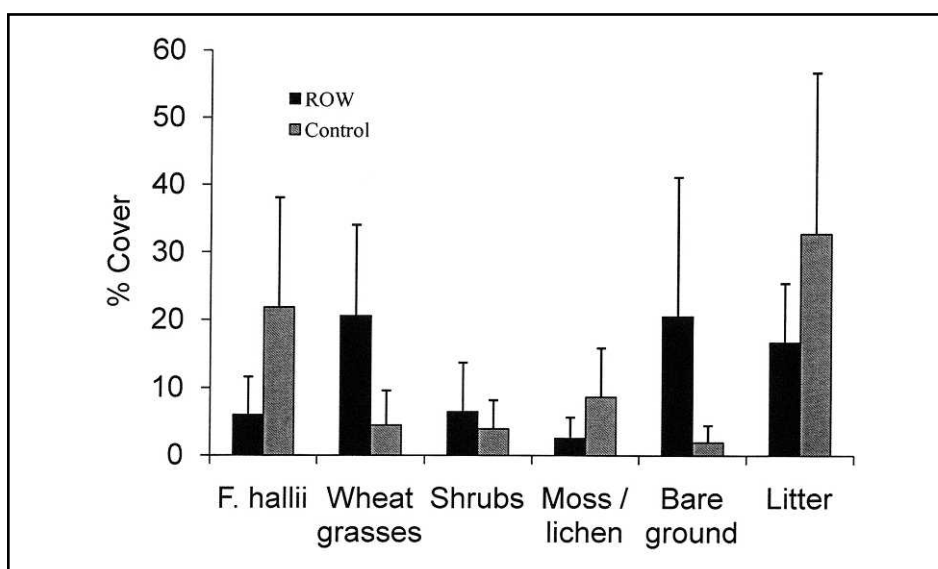


Figure 4. T-test comparison of all control and right-of-way plots showing mean percent cover of *F. hallii* ($P = 0.003$), wheatgrasses ($P < 0.001$), shrubs ($P = 0.279$), moss/lichen ($P = 0.009$), bare ground ($P < 0.001$), and litter ($P = 0.011$). Error bars are standard deviation.

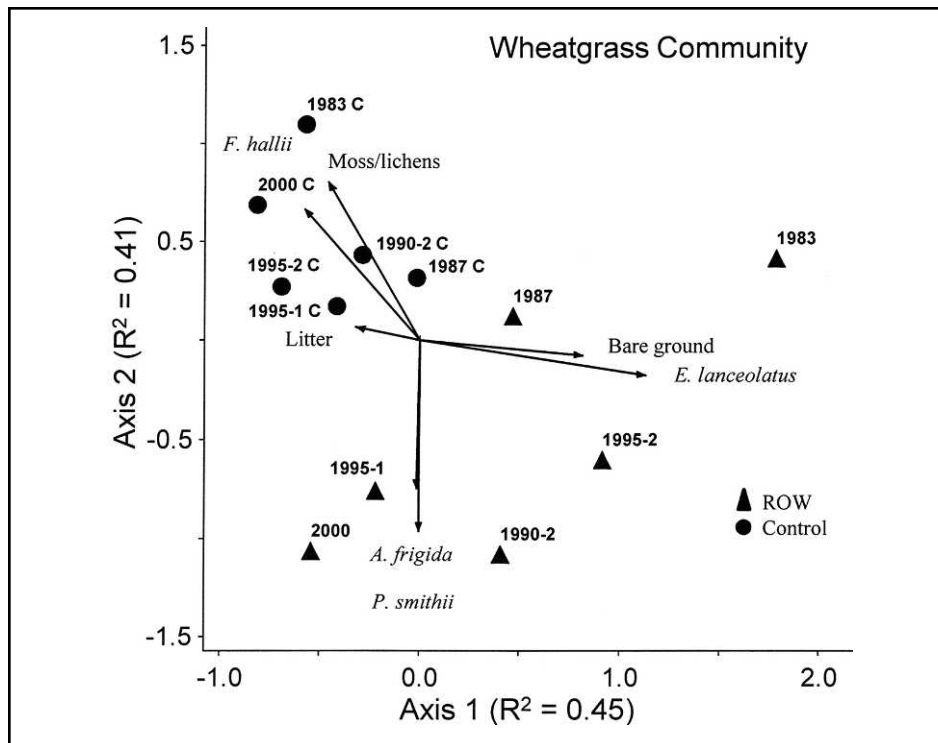


Figure 5. NMS ordination of Wheatgrass community right-of-way and corresponding control plots. Overlays indicate significant trends along axes 1 and 2 indicate significant trends for *F. hallii*, moss and lichens, and litter towards control plots, bare ground, *E. lanceolatus*, and *A. frigida* towards right-of-way plots.

gas well sites than undisturbed controls, and by Ostermann (2001), Desserud et al. (2010), and Elsinger (2009), who found bare ground persisted on pipelines in central and southern Alberta. Bare ground indicates little biological crust, an important component of grasslands. Biological crusts consist of microorganisms (e.g., algae, cyanobacteria) and nonvascular plants (e.g., mosses and lichens) that grow on or just below the soil surface (Pellant et al. 2005). They are important as cover and in stabilizing soil (Belnap and Gillette 1998), and may increase or reduce water infiltration or enhance soil water retention (Pellant et al. 2005). Moss and lichen cover was sparse on ROWs, similar to what Petherbridge (2000) reported on a pipeline work zone shortly after construction. Complete elimination of mosses was reported from top-soil-stripped ROWs in Solonchik and Chernozemic mixed grass prairie (Naeth 1985; Petherbridge 2000; Soulodre 2001). Ostermann (2001) reported the same for a top-soil-stripped

12-year old pipeline in mixed grass prairie. In our study, moss and lichen disappeared from most disturbed sites; and, as Evans and Bishop (1999) concluded, they require many years to recover.

Opinions differ on time required to achieve natural recovery. Returning to the original ecosystem is typically a slow process, possibly not accomplished within a human lifetime (Allen 1993). It may take 50 – 300 years (Dobson et al. 1997) or over 3000 years in places like the Mojave Desert (Lovich and Bainbridge 1999). In western Canadian grasslands, Nasen (2009) found impacts on soil and vegetation by oil and gas development persisted over 50 years, while Naeth (1985) concluded there was evidence of succession towards undisturbed vegetation on a pipeline after 26 years. Our pipelines showed evidence of recovery, with plant communities similar to original grassland within 2 to 23 years. Nevertheless, the majority of our sites lacked a key feature of grassland ecosystems – biological crust – as bare ground remained.

CONCLUSIONS AND MANAGEMENT RECOMMENDATIONS

These results confirm the importance of minimum disturbance in grasslands. Intact sod reduces soil exposure, preventing invasion by rhizomatous species and allowing other native species to propagate the narrow disturbance through seed rain or existing propagules. Rough fescue appears to recover better on plow-in pipeline ROWs than from seeding, most likely from intact sod. Retaining grassland sod through plow-in and keeping disturbance as narrow as possible is critical to successful restoration. Narrow trenching with plow-in pipelining techniques is recommended for grasslands.

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Peggy A. Desserud recently earned a PhD in the department of Renewable Resources, University of Alberta, Edmonton, AB. She is continuing her research as a post-doctoral fellow at the University of Calgary, Calgary, AB, focusing on grassland restoration, plant ecology, plant and soil microbial interactions, and plant community development.

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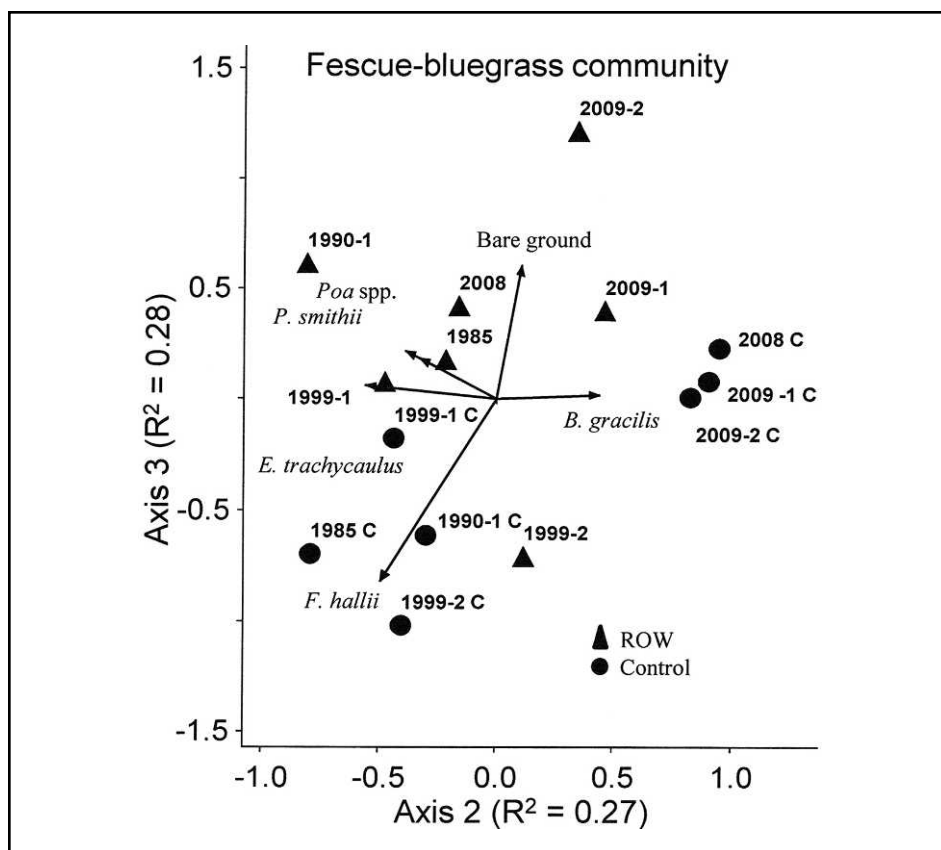


Figure 6. NMS ordination of Fescue-Bluegrass community right-of-way and corresponding control plots. Overlays indicate significant trends along axes 2 and 3, indicate significant trends for *B. gracilis* towards control plots, bare ground, *P. smithii* and *Poa* spp. towards right-of-way plots, and *F. hallii* and *E. trachycaulus* towards both right of way and controls.

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